DC Motor Modeling – Transfer Functions

1) Model and Governing Equations

We consider a separately excited DC motor with armature voltage *v(t)*, current *i(t)*, angular velocity $\omega(t)$, and angular position $\theta(t)$.
Electrical dynamics:

$$v(t) = L\, di/dt + R\, i(t) + K_{\blacksquare}\, \omega(t)$$

Mechanical dynamics:

$$J\, d\omega/dt + b\, \omega(t) = \tau(t),\ \tau(t) = K_{\blacksquare}\, i(t)$$

2) Laplace-Domain Relations

Transforming to Laplace domain (zero initial conditions):

$$V(s) = L\, s\, I(s) + R\, I(s) + K_{\blacksquare}\, \Omega(s)$$

$$J\, s\, \Omega(s) + b\, \Omega(s) = K_{\blacksquare}\, I(s)$$

3) Transfer Function: Voltage → Angular Velocity

After algebraic manipulation:

$$\Omega(s)/V(s) = K_{\blacksquare} / [\, (J\, s + b)(L\, s + R) + K_{\blacksquare}K_{\blacksquare}\,]$$

4) Transfer Function: Voltage → Position

Since $\Theta(s) = \Omega(s)/s$:

$$\Theta(s)/V(s) = K_{\blacksquare} / [\, s((J\, s + b)(L\, s + R) + K_{\blacksquare}K_{\blacksquare})\,]$$

5) Numerical Example

For parameters J=0.01, b=0.1, K■=K■=0.01, R=1 Ω, L=0.5 H, the velocity transfer function is:

$$G\omega(s) = 0.01 / (0.005\, s^2 + 0.06\, s + 0.1001)$$

$$\approx 2 / (s^2 + 12\, s + 20.02)$$

Corresponding position transfer function:

$$G\theta(s) = 2 / [\, s\, (s^2 + 12\, s + 20.02)\,]$$